

WILKES-BARRE SCRANTON, PA, USA

WMO: 725130

Lat: 41.334N

Lon: 75.727W

Elev: 284

StdP: 97.97

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14777

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.8	-22.1	0.5	-14.2	-19.6	0.7	-11.6	10.8	-0.6	9.5	-2.0	3.7	230	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.8	22.3	30.3	21.5	28.8	20.7	23.9	29.0	23.0	27.8	22.2	26.7	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.6	26.2	21.5	16.7	25.3	20.7	15.9	24.6	72.7	29.0	69.3	27.9	66.4	26.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	7.1	DB	-18.6	34.1	3.4	1.6	-21.1	35.2	-23.1	36.1	-25.0	37.0	-27.5	38.2	(4)
(5)			WB	-19.2	25.5	3.3	1.2	-21.6	26.4	-23.5	27.1	-25.3	27.8	-27.7	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-2.9	-1.4	3.0	9.7	15.5	20.0	22.5	21.5	17.7	11.4	5.5	0.3	(6)
(7)		DBStd	10.01	6.17	5.53	5.66	5.13	4.37	3.49	2.92	2.88	3.93	4.59	4.98	5.12	(7)
(8)		HDD10.0	1517	403	321	228	66	5	0	0	0	1	39	152	303	(8)
(9)		HDD18.3	3314	659	553	475	262	111	23	3	6	58	219	386	558	(9)
(10)		CDD10.0	1627	2	1	12	58	176	299	388	357	233	82	17	3	(10)
(11)		CDD18.3	382	0	0	1	4	24	73	133	104	39	4	0	0	(11)
(12)		CDH23.3	3088	0	0	4	56	242	582	1112	781	287	23	1	0	(12)
(13)		CDH26.7	893	0	0	1	13	64	156	365	218	74	2	0	0	(13)
(14)	Wind	WSAvg	2.9	3.3	3.4	3.5	3.4	2.8	2.5	2.4	2.2	2.4	2.6	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	946	59	54	67	78	88	98	93	89	97	79	75	66	(15)
(16)		PrecMax	1552	165	205	138	243	204	232	184	269	265	206	195	167	(16)
(17)		PrecMin	666	10	8	12	24	21	7	26	31	18	1	20	22	(17)
(18)		PrecStd	187	34	33	28	39	39	47	42	44	63	46	35	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	16.4	22.6	28.6	30.8	32.4	34.0	33.2	31.6	26.1	20.7	17.3	(19)
MCWB			13.8	12.1	14.5	16.7	20.6	23.0	23.2	23.1	21.2	19.2	14.7	14.2	(20)	
2%		DB	12.2	12.0	18.2	24.5	28.6	30.3	31.9	30.9	28.7	23.2	17.8	13.2	(21)	
		MCWB	9.6	8.0	11.3	14.9	19.2	21.9	22.6	21.7	20.1	17.5	13.7	10.9	(22)	
5%		DB	8.4	9.0	14.9	21.3	26.1	28.4	30.4	29.0	26.6	21.2	15.6	10.1	(23)	
		MCWB	6.1	5.5	9.6	13.6	18.1	20.8	21.9	21.0	19.6	16.5	11.9	7.9	(24)	
10%		DB	5.4	6.3	11.9	18.6	23.6	26.8	28.7	27.5	24.5	19.1	13.4	7.6	(25)	
		MCWB	3.5	3.5	7.7	11.8	16.8	19.7	21.1	20.3	18.7	14.9	10.0	5.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	13.1	15.3	18.8	22.4	24.3	25.5	24.9	23.4	20.8	16.7	14.5	(27)
MCDWB			16.2	15.3	20.9	24.3	28.3	29.8	31.4	30.1	28.2	23.2	19.1	16.9	(28)	
2%		WB	10.5	8.6	12.7	16.6	20.7	23.1	24.0	23.4	21.9	19.0	14.7	11.2	(29)	
		MCDWB	11.8	11.2	16.7	22.5	26.2	28.3	29.4	28.0	26.3	22.1	17.3	12.6	(30)	
5%		WB	6.4	5.8	10.4	14.8	19.4	22.1	23.0	22.5	20.9	17.4	12.5	8.3	(31)	
		MCDWB	8.3	8.4	14.3	19.9	24.4	26.5	28.0	26.8	24.3	20.2	15.0	10.0	(32)	
10%		WB	3.8	3.7	7.8	12.8	18.0	21.1	22.3	21.7	19.9	15.3	10.5	5.5	(33)	
		MCDWB	5.2	6.1	11.8	17.8	22.2	25.3	26.8	25.6	23.3	18.4	12.8	7.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	9.5	11.1	11.3	10.7	11.0	10.6	9.8	8.3	6.7	(35)	
MCDBR			10.6	11.6	13.7	15.2	14.4	12.8	12.5	12.3	13.1	12.4	11.2	9.5	(36)	
MCWBR			9.3	8.9	8.6	8.3	6.7	5.8	5.0	5.1	5.9	6.9	7.8	8.1	(37)	
5% WB		MCDBR	10.1	10.6	12.2	13.2	12.3	10.7	10.5	10.1	10.1	10.2	9.7	9.1	(38)	
		MCWBR	9.3	9.1	9.0	8.4	6.6	5.6	4.8	4.7	5.3	6.8	7.8	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.326	0.351	0.372	0.409	0.462	0.467	0.455	0.374	0.362	0.316	0.298	(40)	
taud		2.401	2.317	2.350	2.345	2.273	2.152	2.152	2.174	2.377	2.414	2.504	2.481	(41)		
Ebn at Noon		851	878	896	898	869	820	811	809	860	828	831	828	(42)		
Edh at Noon		81	102	111	120	133	150	149	141	107	93	73	69	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.64	2.50	3.48	4.51	5.13	5.53	5.73	4.99	4.09	2.74	1.86	1.35	(44)	
(45)		RadStd	0.12	0.31	0.30	0.49	0.54	0.51	0.35	0.31	0.51	0.33	0.21	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-168	+155	+75		
(47)	Regional (53 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+63		

Nomenclature: See separate page